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SpectralMamba: Computationally Efficient Mamba for Hyperspectral Image Classification

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***Abstract*—Recurrent neural networks and Transformers have recently dominated most applications in hyperspectral (HS) imaging, owing to their capability to capture long-range dependencies from spectrum sequences. However, despite the success of these sequential architectures, the non-ignorable inefficiency caused by either difficulty in parallelization or computationally prohibitive attention still hinders their practicality, especially for large-scale observation in remote sensing scenarios. To address this issue, we herein propose SpectralMamba – a novel state space model incorporated efficient deep learning framework for HS image classification. SpectralMamba features the simplified but adequate modeling of HS data dynamics at two levels. First, in spatial-spectral space, a dynamical mask is learned by efficient convolutions to simultaneously encode spatial regularity and spectral peculiarity, thus attenuating the spectral variability and confusion in discriminative representation learning. Second, the merged spectrum can then be efficiently operated in the hidden state space with all parameters learned input-dependent, yielding selectively focused responses without reliance on redundant attention or imparallelizable recurrence. To explore the room for further computational downsizing, a piece-wise scanning mechanism is employed in-between, transferring approximately continuous spectrum into sequences with squeezed length while maintaining short- and long-term contextual profiles among hundreds of bands. Through extensive experiments on four benchmark HS datasets acquired by satellite-, aircraft-, and UAV-borne imagers, SpectralMamba surprisingly creates promising win-wins from both performance and efficiency perspectives. The code will be available at https://github.com/jingyao16 for the sake of reproducibility after a possible acceptance for publication.**

***Index Terms*—Artificial intelligence, efficient, Mamba, hyperspectral image classification, state space model, spatial-spectral, transformer, remote sensing.**

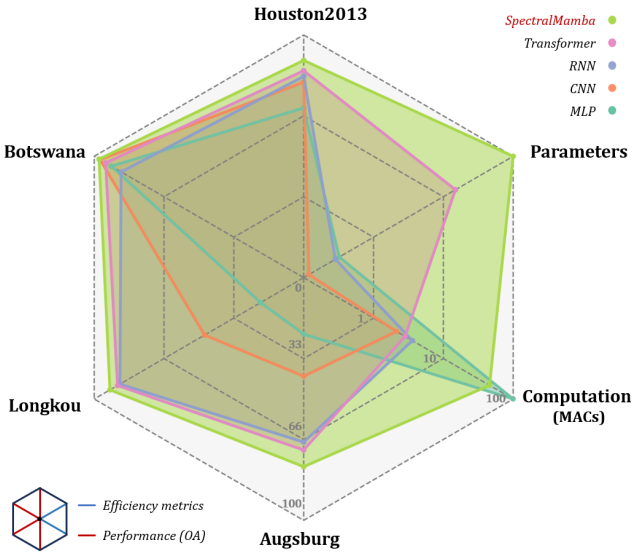


Fig. 1. The radar chart of experimental results of SpectralMamba and classic network architectures in terms of both hyperspectral image classification performance metric (OA) and mean efficiency metrics (number of parameters and MACs) on four benchmark datasets. To better visualize their differences, we set the lowest values of parameter number and MACs as the base score of 100, and customize a base-10 logarithmic scale on the MACs-axis. According to the chart, our SpectralMamba significantly outperforms its competitors along most metrics, showcasing its great potential as a novel efficient and effective deep learning framework for hyperspectral data analysis.

I. Introduction

THE emergent development of hyperspectral (HS) imaging remarkably empowers humans in observing the real world in greater detail and depth [1]. Unlike traditional photography which acquires images in a limited number of broad spectral bands, the HS imaging system unprecedentedly achieves the simultaneous capture of both spatial and spectral information by measuring a spectrum of the energy at each pixel. The generated 3-dimensional (3-D) HS data cube contains the near-continuous spectral profile for each spatial resolution element, thus allowing for more accurate quantification, identification, and recognition of the imaged contents. Owing to the recent progress in aerospace and instrumental technology, HS imaging has gradually become an indispensable tool for remote sensing (RS) [2]. Of all the widespread applications it has found, HS image classification has drawn considerable attention across diverse fields, from environmental monitoring [3] to urban planning and military science [4], demonstrating its pervasive potential and cross-cutting importance [5].

The primary objective of HS image classification in RS is to accurately identify the various land cover or land use types of interest within the image by leveraging the detailed spectral signatures associated with each pixel. Despite the capability of HS imaging to capture hundreds of narrow wavelength bands (typically ranging from the visible spectrum to the

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near-infrared region), providing in-depth characterization of the spectral properties of different materials, there remain two long-standing challenges in its practical applications.

1) The curse of dimensionality, also known as the Hughes phenomenon, is often encountered in HS image classification when the classification accuracy undergoes an initial rise with more spectral bands observed but then decreases dramatically once a certain number of bands is reached [6]. The fundamental cause of this issue stems from the exponential growth of the feature space volume as the number of dimensions expands, rendering the computational processing and effective analysis of the HS data increasingly burdensome and challenging.
2) Spectral variability and spectral confusion are the other two prevalent phenomena that frequently manifest in HS data. The former issue refers to the situation where the same material displays varying spectral characteristics under different conditions, such as changes in illumination, atmospheric effects, or intrinsic variations, while the latter occurs when distinct materials exhibit similar spectral profiles [7].

In addition, several other issues always arise in conjunction with these challenges in HS data analysis, such as the limited availability of labeled training samples, and inevitable sensor noise with complex distributions, which makes it more challenging to precisely distinguish ground objects based solely on their spectral reflectances.

The field of RS has dedicated significant efforts to tackle these challenges from either a single or multiple perspectives over the past decades. In the early stages, they commonly resort to statistical approaches to achieve simultaneous dimensionality reduction and feature extraction. Besides the primarily adopted principal component analysis and independent component analysis, the ever-advancing machine learning approaches open up new possibilities for representing HS data in a more compact and discriminative form. A successful approach is based on kernel tricks that typically map HS images from original complex data space into a latent feature space with higher dimensionality [8]. Additionally, some other researcher also investigated linear discriminant analysis [9], subspace learning [10], manifold learning by solving an explicitly regularized loss function, and heuristics techniques encompassing ensemble methods [11], and active learning strategies [12], to effectively process and analyze HS data for precise pixelwise recognition. During this period, shallow machine learning models, such as nearest neighbors, decision trees, and support vector machines, emerged as prevalent choices for effective backend classifiers complementing these feature extraction methods.

As deep learning (DL) has proliferated across numerous research fields since the last decade, the RS community has also embraced this powerful learning paradigm for HS data analysis, harnessing its capability to learn representations directly from the data, thereby mitigating the intrinsic cognitive bias imposed by inadequate mathematical modeling in conventional approaches [13]–[16]. Among the various DL architectures for HS image classification, convolution neural network (CNN) has taken pride of place over a long period. Benefiting from the local receptive ability through shift-invariant convolutions, typical CNNs successfully realize multi-level–from low-level to high-level–feature extraction and semantic abstraction along spatial and spectral dimensions through end-to-end training from input-target pairs. However, although this family of models is good at exploiting local contextual information, their inherent local connectivity and weight sharing inevitably restrict the modeling of long-range correlations and dynamics within and across data sequences, respectively.

At this point, sequence models, such as recurrent neural networks (RNNs) and Transformers, came into the public eye for their effectiveness in processing high-dimensional sequential data. By orderly unfolding the HS spectrum into a long sequence, RNNs and Transformers essentially enable the capture of long- and short-term spectral fingerprints through recurrent state modeling and attention mechanism, respectively. Besides, these sequence models have been widely proven to be more competent in handling non-linear data dynamics under complex RS scenes than CNNs. Despite these merits, they inherently suffer from parallel training difficulty and burdensome pairwise multiplicative computations, respectively. Although enormous efforts have been made to cope with these issues, as of yet, most of these variants can hardly avoid becoming increasingly sophisticated in either network structures or working flows in pursuing accuracy breakthroughs with improved representation, appearing to have reached a plateau that cannot avoid trading-off between performance and computational efficiency.

Fortunately, recent advances in the state space model (SSM) make it broadly applicable and provide a novel avenue for sequentiality modeling. Building upon the theoretical foundation of classical SSM from control theory and powerful modern DL advantages, the emerging deep SSMs unprecedentedly allow for efficient computation in learning very long-range dependencies over tens of thousands of timesteps and are dominating more and more benchmarks across various domains [17]. Nevertheless, existing SSMs are typically designed for causal learning on low-dimensional sequences such as audio and language, leaving their practicality on high-dimensional visual data such as HS imagery underexplored [18]. Therefore, in this work, we excavate the potential of tailoring SSM to HS data by taking a deep dive into their traits. To be more specific, we propose SpectralMamba – an efficient and effective SSM-integrated DL framework for both pixelwise and patchwise input-based HS image classification. SpectralMamba leverages a simplified but adequate modeling of HS data dynamics in both spatial-spectral feature space and hidden state space, thereby alleviating the effect caused by spectral variability and spectral confusion. The underlying computational overhead caused by parameter size and computations is further reduced by a customized scanning strategy that can additionally enhance sequential representation while maintaining local spectral fingerprints of HS data. The main contributions of this article can be highlighted as follows.

1) We propose a novel SSM-based backbone network termed SpectralMamba that makes a further step towards performant and computation-friendly HS image classi-

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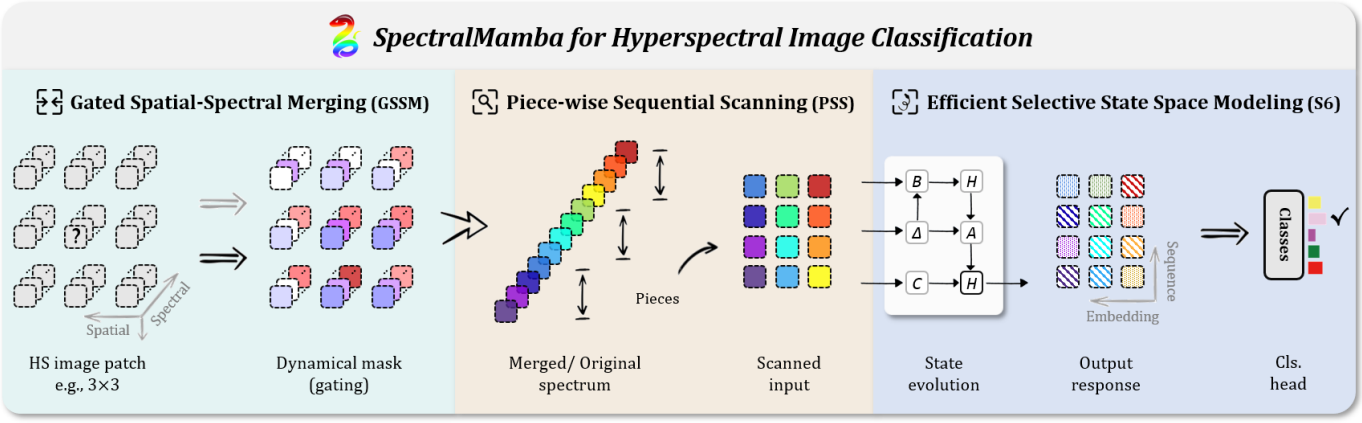


Fig. 2. SpectralMamba mainly consists of three components, i.e., gated spatial-spectral merging (GSSM) module, piece-wise sequential scanning (PSS) strategy, and efficient selective state space (S6) modeling. Its pixelwise counterpart functions by directly operating the original spectrum from the middle stage. A, B, C, and Δ denote all learnable parameters in the hidden state space, where H records the selectively embedded sequence for the final output.

fication from the perspective of sequence modeling. To the best of our knowledge, this is the first work that well tailors the deep SSM for HS data and its analysis.

2) Targeting the high dimensionality of HS data, and the issues of spectral variability and confusion, we propose the strategies of piece-wise sequential scanning (PSS) and gated spatial-spectral merging (GSSM) to fully encode the underlying spatial regularity and spectral peculiarity, yielding a more robust discriminative representation learned via a fully lightweight architecture.
3) Through extensive experimental comparison on four benchmark HS datasets acquired from satellite-, aircraft-, and UAV-borne platforms, our SpectralMamba significantly outperforms the representative competitors with classic backbones at a generally minimal computational resource cost (shown in Fig. 1). The ablation studies further verify the effectivity of our key components, such as PSS maximally brings approximately 4% improvement in OA while reducing 60% parameters and 40% computations than our baseline.

The remainder of this article is organized as follows. Section II discusses related work majored along conventional CNN-based and recent sequential models. The preliminary elements for state space models, design of our SpectralMamba, and method analysis were elaborated in Section III. Section IV details the experiments, including descriptions of the datasets and implementation, evaluation of both performance and computational cost, comparison results and analysis, and ablation studies. Finally, Section V concludes the work and points out plausible future directions.

II. Related Work

A. CNN-based Methods for HS Image Classification

Tremendous efforts have been made to build novel HS image classification methods on CNN-based backbones. Owing to its local perceptual capability, the early-stage methods mainly explored the applicability of multiple-dimensional convolutional structures for the HS image classification task. The styles of stacking and multi-stream fusion were typically adopted to fully exploit the underlying spatial and spectral information. After then, transformation-based techniques were widely studied, Wang et al. considered integrating the concept of multi-scale learning into CNNs to allow for a more comprehensive representation of the HS data, improving the model's ability to discern subtle differences between classes [19]. An enhanced 3-dimensional discrete wavelet transform method was introduced by Cao et al. to boost the feature quality by alleviating the noise effect. Wu et al. presented cross-channel CNNs to extract more effective spectral and spatial details [20]. Autoencoder and fully convolutional network architectures have also been widely used to encode more informative patterns by enforcing the self-reconstruction of HS data, for example, Ahmad et al. employed a 3-D convolutional autoencoder to simultaneously extract spectral and spatial features in an unsupervised manner [21]. Yao et al. and Zhang et al. utilized the local and global prior knowledge inherent in superpixel segmentations and connected graphs under the semi-supervised learning framework respectively [22], [23]. Later, Zheng et al. leveraged the unique spectral characteristics of HS images by incorporating the attention mechanism into conventional convolutions to address the challenges posed by rotation and orientation variations in the data [24]. Although recent CNN-based methods have achieved great advancement in enhancing the quality of spatial-spectral feature extraction and the fusion of complementary information, they are still prone to exhibit complex network architectures with high computational costs, and inefficient modeling of spectral sequential properties, leaving room for further improvement on practicability.

B. Sequence Models for HS Image Classification

By treating the spectrum as sequential time-steps, RNNs were naturally introduced for the sequential modeling of HS data. At first, Wu et al. attempted to directly use RNNs in feature extraction of HS data along spectral dimension [25], which fixed the issue of vanishing gradient to some extent. The long short-term memory (LSTM) and its gated counterpart –

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gated recurrent units (GRUs), were then proposed for more efficient spatial-spectral feature extraction [26]. On the basis of these two variants, Hang et al. presented a cascaded version with multiple fusion strategies from feature fusion to decision fusion [27]. The bi-directional RNN was first introduced by Liu et al. [28] and then developed by Liang et al. [29], with more mining of detailed structure information and alleviation of training difficulty using single-stream RNNs. The multi-scale and residual learning strategies were also introduced into GRU to better gain short and long-range correlations [30]. Although these efforts exhaustively explored the practicality of RNN in HS data analysis, there remain some issues not well addressed, such as the inefficient modeling of the lengthy spectrum and joint spatial-spectral dynamics.

The rapid development of Transformer models in recent years has brought new opportunities for analyzing HS data from a sequence perspective [31]. By leveraging the self-attention mechanism to establish long-range dependencies among spectral bands, these methods allow for a more nuanced understanding of the intricate patterns within HS data. He et al. first introduced this architecture into the HS image classification task using a pretrained VGG-like network for visual feature extraction before a spatial-spectral tokenization. Hong et al. then developed a groupwise spatial-spectral feature embedder and a cross-layer connected Transformer encoder, making it customized for the HS data structure for the first time [32]. Zou et al. addressed the issue of inaccurate token embedding by carefully devising an HS embedder and a local-enhanced Transformer encoder [33]. More recent works focus on either more advanced backbones such as 3-D swin Transformer or more efficient training strategies like masked self-pretraining and contrastive learning [34], [35]. However, challenges such as the need for large labeled datasets, the difficulty in fully exploiting local spatial-spectral information, and the unbearable computational complexity still exist, which necessitates further research on exploring a more performant and efficient solution.

III. Methodology

A. Preliminaries

1) State Space Model: Inspired from classical SSMs [36] and modern DL advances, most notably CNNs, RNNs, and Transformers, structured state space sequence models (S4) [17], [37] have recently emerged for modeling sequential data. This class of models typically originates from a continuous-time system that maps an input function or sequence $x(t) \in \mathbb{R}^M$ to an output response signal $y(t) \in \mathbb{R}^O$ through an implicit latent state $h(t) \in \mathbb{R}^N$, which can be mathematically formulated using the following ordinary differential equations,

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \tag{1}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t), \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ and $\mathbf{C} \in \mathbb{R}^{O\times N}$ control how the current state evolves over time and translates to the output, $\mathbf{B} \in \mathbb{R}^{N\times M}$ and $\mathbf{D} \in \mathbb{R}^{O\times M}$ depict how the input influences the state and the output, respectively. Herein we consider the case of a single-input single-output system with $O = M = 1$ and omit $\mathbf{D}x(t)$ term by treating it as a skip connection as the S4 model does [17].

2) Discretization: The first step in applying SSMs to discrete signals such as language, audio, and images, is to transform the system parameters into their "discretized" counterpart. A commonly adopted discretization method is the zero-order hold rule, by which the reparameterization holds as follows,

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \tag{3}$$

$$\begin{aligned}\bar{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\bar{\mathbf{A}} - \mathbf{I})(\Delta\mathbf{B}) \\ &\approx (\Delta\mathbf{A})^{-1}(\Delta\mathbf{A})(\Delta\mathbf{B}) \\ &= \Delta\mathbf{B},\end{aligned} \tag{4}$$

where the first-order Taylor series approximation is used by following [38]. The timescale parameter Δ denotes the sampling step, i.e., $x_k = x(k\Delta)$, which also trades off the state and current input during the evolution. Then the discrete SSM can be formulated into the following recurrent representation,

$$h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k, \tag{5}$$

$$y_k = \mathbf{C}h_k, \tag{6}$$

which can be computed similarly to RNNs. To better accommodate GPU acceleration for efficient training, S4 also unrolls the above linear recurrence, yielding its global convolutional representation as

$$\mathbf{y} = \mathbf{x} * \bar{\mathbf{K}}, \tag{7}$$

where $\bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}})$ represents the SSM convolution kernel, and L is the sequence length of input.

3) Mamba: Besides the linearity, another simplifying assumption of the above systems is the time invariance, that is, all the system parameters are defined as time-independent. Recently, a novel class of selective SSMs (S6) breaks this constraint by parameterizing $(\Delta, \mathbf{B}, \mathbf{C})$ as functions of input $\mathbf{x}$, thus endowing SSMs with additional selection ability to focus on the important or ignore the unimportant. As a common fashion, a simplified neural network architecture consists of linear layers, a convolution layer, a residual connection, nonlinear transformations, and most importantly the S6 kernel together to form the Mamba block. A hardware-aware optimization is also proposed to guarantee its efficient implementation.

4) Scanning: In the context of visual encoding, scanning lies as a primary step before feeding into Mamba models. The key of scanning is to convert high-dimensional inputs into a sequence format that can be processed by the Mamba block according to specific rules. Early attempts, such as Vision Mamba [39] and ChangeMamba [40], mostly adopted a omni-directional, multiple sequential scan style to record information from multile perspectives, hoping to fully capture the sequential properties of the data. However, this vanilla and potentially redundant operation has been gradually optimized in subsequent research. Besides MS-VMamba [41], which introduced a multi-scale strategy, other efficient scanning strategies have emerged, such as LocalVMamba [42], which proposed using scans based on local windows; the Stripe-shift Mechanism [43] that obtains sequence encodings through a

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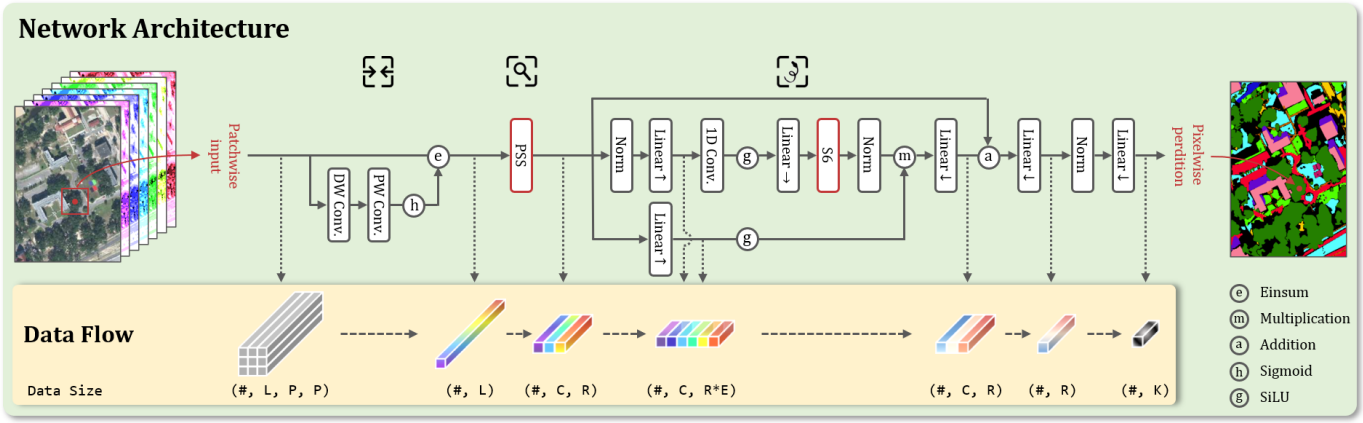


Fig. 3. Detailed architectural design and data processing pipeline of our proposed SpectralMamba, exemplified using a patchwise input under batch training for hyperspectral image classification, where PSS and S6 refer to the piece-wise sequential scanning and selective state space model, respectively.

nested S-shape scan pattern; and the attention-based reshuffling pattern scan [44], focusing only on semantically attentive regions while preserving spatial integrity. Despite these advancements, customized scanning strategies specifically for HS data remain relatively scarce.

B. SpectralMamba: Overview

To break through the performance and efficiency bottlenecks of existing methods based on CNN, RNN, or Transformer backbones, we propose the SpectralMamba, a Mamba – S6 model incorporated DL solution to tackle HS image classification. The key of SpectralMamba lies in its simultaneous modeling of HS data dynamics by gated spectrum merging in the spatial-spectral space and selective sequence learning in hidden state space through a minimally parameterized network architecture. Furthermore, a novel sequential scanning strategy tailored for HS data is proposed to make the framework more computation-friendly by unraveling the hundreds-of-bands spectrum into pieces. The patchwise SpectralMamba can also be flexibly transformed into its pixelwise counterpart by skipping the spatial-spectral encoding stem. Fig. 2 gives an illustration of how our proposed SpectralMamba works, and the detailed network architecture as well as its data flow are shown in Fig. 3.

C. SpectralMamba: Key Components

Let the 1-D vector $\mathbf{x}^{pixel} = [\mathbf{x}_1^{pixel}, \ldots, \mathbf{x}_L^{pixel}] \in \mathbb{R}^{1\times L}$ denote a given pixel of an HS image, where L is the band number of the spectrum. We consider the state space modeling within the spectral domain, that is, our aim becomes to find its output response $\mathbf{y}^{pixel} \in \mathbb{R}^L$ through a well-defined S6 model. However, simply treating the reflectance values at each band as the representation may limit the mining of sequential patterns. Therefore we expand the model dimension by a factor, $E = 8$ in our case, to enlarge the state space size. As the core architectural part in Fig. 3 depicts, the Mamba block designed for HS data consists of three streams. Its mainstream comprises two distinct LayerNorm layers in input and expanded state spaces, three linear layers for expanding, keeping, and compressing feature dimensions, a SiLU nonlinear activation g, and the S6 block. The other two streams are the skip connection and an excitation-like multiplication to adaptively transform original information across layers [45]. Note that the use of skip connection and nonlinearity is crucial for stable training with fast convergence, while the practical performance appears to be not that sensitive to the choice of normalization and activation function.

1) Piece-wise Sequential Scanning: The aforementioned state space modeling is empowered to pay attention or forget features at particular wavelengths through an input-dependent parameterizing way [38]. When applied to HS data with hundreds of near-continuous bands, its high spectral redundancy drives us to rethink the input manner. Unlike the very recent efforts in modifying Mamba for natural image processing by considering spatial multi-directional scanning [46], we propose a novel piece-wise sequential scanning (PSS) along the spectral dimension to fully leverage the reflectance characteristics of different types of ground objects.

In specific, we can formulate the PSS module as

$$PSS(\mathbf{x}^{pixel}) = [S_1\mathbf{x}^{pixel}, \ldots, S_R\mathbf{x}^{pixel}], \tag{8}$$

where $S_r\mathbf{x}^{pixel} = [\mathbf{x}^{pixel}_{(r-1)C+1}, \ldots, \mathbf{x}^{pixel}_{rC}]^\top \in \mathbb{R}^{C\times 1}$ scans continuous pieces from the original spectrum $\mathbf{x}^{pixel}$ for $r = 1, \ldots, R$, and R is the number of pieces with length C. By design, the pieces are contiguous and non-overlapping with equal length $C = L/R$, where piece r covers bands $\{(r-1)C+1, \ldots, rC\}$. When L is not exactly divisible by R, the spectrum is zero-padded to the nearest multiple. This simple uniform partitioning maintains generality across different sensors without requiring prior knowledge of band centers or spectral response functions, while allowing the subsequent input-dependent S6 model to learn which spectral sub-regions are most informative. This also acts similarly as a resampling with features at each position in the sequence enriched. By applying PSS before our Mamba block, the response accordingly turns from a 1-D L-length vector to a 2-D output of shape $C \times R$. We then add one more pre-layer before the common softmax-based classification head to finally obtain K-length categorical logits.

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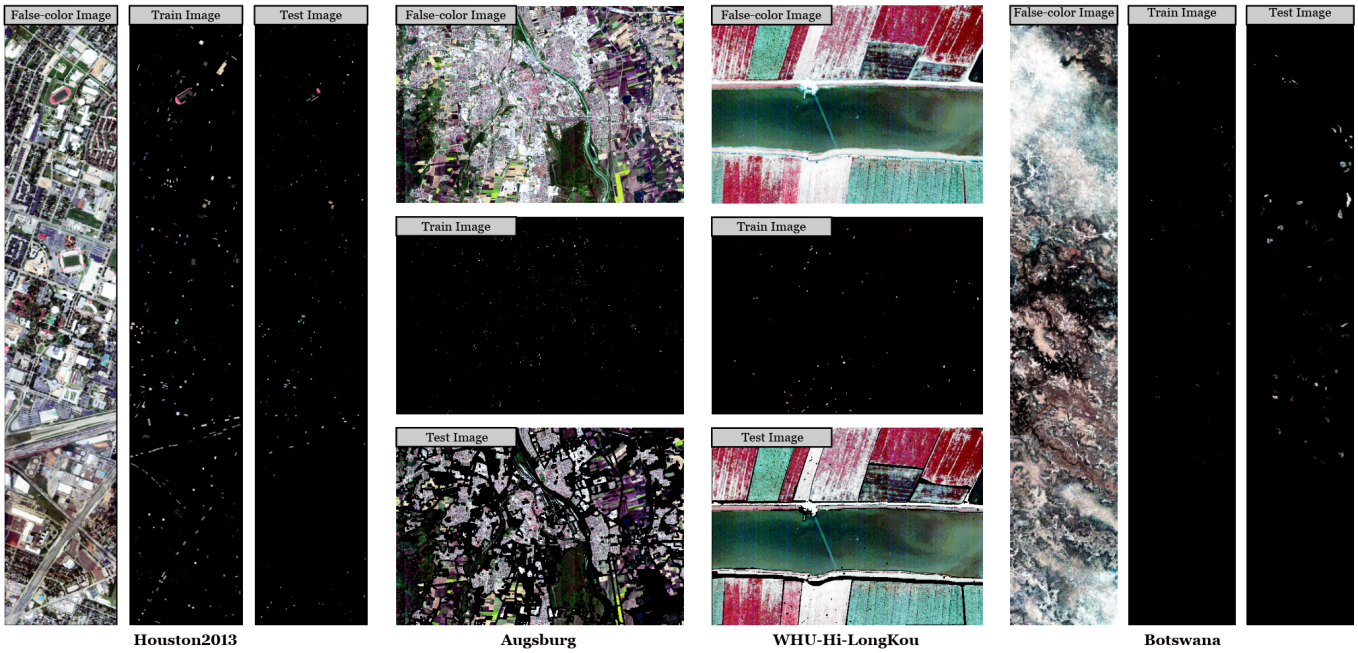


Fig. 4. An illustration of the four hyperspectral image classification datasets investigated in this work. The train and test images on the first three were obtained using our proposed sample split strategy.

2) Gated Spatial-Spectral Merging: It is crucially important to consider spatial information for discriminative representation learning. However, conventional spatial-spectral feature extraction methods commonly treat each patch equally with fixed convolution kernels. Inspired by the S6 model that learns interactions along the sequence in an input-dependent way, we propose to further increase the content-awareness by introducing a dynamical gate function for adaptive spatial-spectral embedding. With the proposed gated spatial-spectral merging (GSSM), we can substitute the $\mathbf{x}^{pixel}$ in Eq. (8) with merged spectrum computed as follows,

$$GSSM(\mathbf{x}^{patch}) = h(f_{PW}(f_{DW}(\mathbf{x}^{patch}))) \otimes \mathbf{x}^{patch}, \quad (9)$$

where h is the sigmoid activation, f represents the composite function composed of depthwise (DW) convolution and pointwise (PW) convolution, $\otimes$ denotes the Einstein summation along the spatial dimension that combines two tensors of shape $L \times P \times P$ into a 1-D vector with length L. More specifically, given a patchwise input $\mathbf{x}^{patch} \in \mathbb{R}^{L \times P \times P}$ (where $P = 3$), the DW convolution f_{DW} operates independently on each of the L channels with a 3×3 kernel (stride=1, padding=1, no dilation), preserving the spatial resolution with output shape $\mathbb{R}^{L \times P \times P}$. The PW convolution f_{PW} then applies a 1×1 kernel to mix information across the L channels. The sigmoid activation h produces a dynamic mask $\mathbf{M} \in \mathbb{R}^{L \times P \times P}$ with values in $[0, 1]$. The Einstein summation $\otimes$ performs weighted spatial aggregation via einsum$(lhw, lhw \rightarrow l)$, which element-wise multiplies $\mathbf{M}$ with $\mathbf{x}^{patch}$ and sums over the spatial dimensions, yielding a merged 1-D spectrum $\in \mathbb{R}^{L}$. Through GSSM, we hope to adaptively encode the semantic relationships between the center pixel and its neighborhoods in learning a more discriminative "spectrum", thereby attenuating spectral variability and confusion effect.

D. SpectralMamba: Method Analysis

It is non-trivial to directly extend SSM to applications of HS data. Building upon the insights on SSM and structural prior knowledge of HS data, our SpectralMamba offers a viable SSM-based baseline to address the dense prediction application of HS images. The proposed PSS strategy not only enables the model to uncover local characteristics of the spectral profile, but also further improves efficiency by narrowing the width of core operation networks. From the perspective of efficiency, PSS partitions the original L-length spectrum into R shorter pieces of length $C = L/R$, directly reducing the sequential state evolution cost in S6. From the perspective of performance, HS data exhibit strong local spectral continuity among adjacent bands; grouping them into pieces preserves this structure within each piece while the piece-level features form a coarser global spectral profile, yielding a dual-scale representation. Moreover, PSS effectively acts as a feature enrichment, transforming each single-valued spectral position into a C-dimensional feature vector, and as an implicit regularizer, where shorter sequences are easier for S6 to learn from limited training samples. Moreover, the GSSM module is designed based on the observation that the semantic relationships between the central pixel and its neighboring pixels usually vary spatially and spectrally across the scene. The widely-existed mixed pixel phenomenon can also differ for pixels within a local patch, especially for those falling near boundaries. We hope to efficiently capture such highly spatial-spectral-variant HS data dynamics through a lightweight mask learner, thus yielding a merged spectrum with increased discriminative ability for the subsequent sequential learning in the state space.

The connections between our proposed SpectralMamba and related works are also worth noting. On one hand, although

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CasRNN – an RNN-based representative for HS image classification – has considered the similar spectral redundancy issue by hierarchically learning from adjacent spectral bands to nonadjacent ones, their imparallelizable recurrence trait still accumulated both the computations and parameters in pursuing a stable training [27]. Also, the unbearable quadratic complexity of self-attention in conventional Transformers has significantly magnified the computational burden as the number of neighboring spectral bands considered in a so-called groupwise embedding increases [32]. In contrast, the proposed PSS in our SpectralMamba perfectly matches the efficient feature selection via state evolution in S6, simultaneously preserving local spectral patterns and enlarging the feature dimension of sequence, while further improving computational efficiency. What's more, our practice verifies that a non-overlapped scanning, i.e., $R = L/C$, is enough to produce a promising performance at a lower computational overhead. On the other hand, different from conventional gated convolutions for natural images [47], our GSSM provides a lightweight gating mechanism in capturing the highly spatial-spectral-variant HS data dynamics. It not only fits the setting to maintain the spectral sequentiality but also complements the subsequent content-aware learning in the state space with efficient spatial-spectral rectification. In the following experiments section, we will demonstrate how SpectralMamba outperforms these predecessors by enhancing the efficacy in interpreting HS data on the abovementioned merits, meanwhile, maintaining high efficiency with low computational resource requirements.

Last but not least, we analyze the computational complexity of our proposed model. As depicted in Figure 3, given an HS image with L spectral bands, the first GSSM module exhibits complexity of $\mathcal{O}(LHW \cdot (L+K^2))$, which mainly comes from the depthwise and pointwise convolutions, where H and W are the height and width respectively, and the kernel size K equals to 3 in our case. The remaining part of the network is mainly composed of multiple linear layers and the S6 module. After PSS, the sequence length reduces from L to $C = L/R$, while the feature dimension increases from 1 to R. The S6 module processes sequences of length C with complexity $\mathcal{O}(C \cdot E \cdot N)$ per pixel, where E and N are the expansion factor and state dimension, respectively. However, the linear layers that expand and compress the feature dimension (from R to ER and back) contribute $\mathcal{O}(C \cdot R^2) = \mathcal{O}(L^2/C)$ per pixel. Aggregating over all HW pixels, the end-to-end complexity is $\mathcal{O}(L^2HW/C + L^2HW) = \mathcal{O}(L^2HW)$. Notably, although the asymptotic order remains $\mathcal{O}(L^2HW)$, the practical benefit of PSS is substantial: it shifts computation from sequential state evolution (harder to parallelize) to matrix multiplications in linear layers (highly GPU-friendly), significantly reducing the actual constant factor as verified by our MACs measurements.

IV. Experiments

A. Datasets

We select four benchmark HS datasets to conduct experiments, covering all types of acquisition platforms, that are, aircraft-borne, unmanned aerial vehicle (UAV)-borne, and satellite-borne, hoping to give a comprehensive and faithful verification of our proposed SpectralMamba. The details are summarized in Table I.

1) Houston2013 HS Dataset: The first HS dataset Houston2013 consists of HS imagery acquired by ITRES CASI-1500 HS imager with 144 spectral bands ranging from 380 nm to 1050 nm. The investigated scene has 349 $\times$ 1905 pixels at a ground sampling distance (GSD) of 10 m and 15 LULC categories, covering the University of Houston campus and the surrounding urban area. It was provided by the IEEE Geoscience and Remote Sensing Society data fusion contest in 2013 and has been widely used in research and competitions related to HS image analysis [48].

2) Augsburg HS Dataset: The second HS dataset Augsburg consists of HS imagery acquired by the HySpex – an airborne imaging spectrometer system operated by the the Remote Sensing Technology Institute of the German Aerospace Center [10]. The pre-processed HS imagery has 180 bands ranging from 400 nm to 2,500 nm with high quality. Our selected sub-region includes 332 $\times$ 485 pixels at a GSD of 30m, covering the city of Augsburg, Germany. The ground reference map comprising seven categories was elaborately produced by manual labeling based on OpenStreetMap product.

3) Longkou HS Dataset: The third Longkou dataset, also known as WHU-Hi-LongKou, is a specific UAV-borne HS dataset acquired in Longkou Town, Hubei, China. This dataset was captured using the Headwall Nano-Hyperspec sensor mounted on a UAV at a flight altitude of 500 meters. We use the opened data that has 550 $\times$ 400 pixels at a GSD of 0.463 m, and 270 spectral bands ranging from 400 nm to 1,000 nm. More than 204 thousand labelings with six crop types and three LULC categories were used for evaluation. This dataset is part of the WHU-Hi dataset collection, which also includes other two datasets HanChuan and HongHu [50].

4) Botswana HS Dataset: The last Botswana dataset is a collection of HS imagery acquired by the NASA EO-1 satellite over the Okavango Delta area of Botswana between 2001 and 2004. This dataset was captured using the well-known Hyperion sensor, which collects original data at a 30-meter spatial resolution in 242 bands covering the 400-2,500 nm portion of the spectrum. After pre-processing by removing uncalibrated and noisy bands, 145 bands over a 1,476 $\times$ 256 subscene are commonly used for classification experiments. The ground reference comprises 14 identified classes representing different land cover types in the investigated region.

B. Experimental Setup

To ensure the reproduction of our experimental results and a faithful verification of the effectiveness of our method, we herein present the necessary details in implementing all compared methods on investigated datasets.

1) Train and Test Set Split: Besides the data quality, the split of the train and test set has a significant effect on assessing the model performance, particularly the deep models. We strive to reason the popularity of the Houston2013 dataset and conclude four criteria for sample selection with easy practicality and wide acceptability.

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TABLE I
SUMMARY OF FOUR INVESTIGATED HS DATASETS, INCLUDING THE TAGS OF DATA ACQUISITION INFORMATION, CATEGORICAL INFORMATION OF GROUND OBJECTS, AND THE CORRESPONDING NUMBERS OF TRAIN AND TEST SAMPLES.

Dataset	Houston2013			Augsburg			Longkou			Botswana		
Sensor	ITRES CASI-1500			HySpex			Headwall Nano-Hyperspec			Hyperion		
Platform	Aircraft-borne			Aircraft-borne			UAV-borne			Satellite-borne		
Loc. & Date	America, 2012			Europe, 2018			Asia, 2018			Africa, 2001		
GSD	10 m			30 m			0.463 m			30 m		
Wavelength	380 nm - 1050 nm			400 nm - 2500 nm			400 nm - 1000 nm			400 nm - 2500 nm		
Data Size	349 × 1905 × 144			332 × 485 × 180			550 × 400 × 270			1476 × 256 × 145		
Class No.	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
1	Healthy Grass	198	1053	Forest	80	13427	Corn	54	34457	Water	24	246
2	Stressed Grass	190	1064	Residential Area	80	30249	Cotton	53	8321	Hippo Grass	20	81
3	Synthetic Grass	192	505	Industrial Area	80	3771	Sesame	54	2977	Floodplain Grasses 1	21	230
4	Tree	188	1056	Low Plants	80	26777	Broad-Leaf Soybean	53	63159	Floodplain Grasses 2	24	191
5	Soil	186	1056	Allotment	80	495	Narrow-Leaf Soybean	54	4097	Reeds	20	249
6	Water	182	143	Commercial Area	80	1565	Rice	50	11804	Riparian	20	249
7	Residential	196	1072	Water	81	1449	Water	53	67003	Fires Car	20	239
8	Commercial	191	1053	-	-	-	Roads and Houses	53	7071	Island Interior	20	183
9	Road	193	1059	-	-	-	Mixed Weed	51	5178	Acacia Woodlands	21	293
10	Highway	191	1036	-	-	-	-	-	-	Acacia Shrub Lands	22	226
11	Railway	181	1054	-	-	-	-	-	-	Acacia Grasslands	20	285
12	Parking Lot 1	192	1041	-	-	-	-	-	-	Short Mopane	22	159
13	Parking Lot 2	184	285	-	-	-	-	-	-	Mixed Mopane	20	248
14	Tennis Court	181	247	-	-	-	-	-	-	Exposes Soils	20	75
15	Running Track	187	473	-	-	-	-	-	-	-	-	-
Total	-	2832	12197	-	561	77733	-	475	204067	-	294	2954

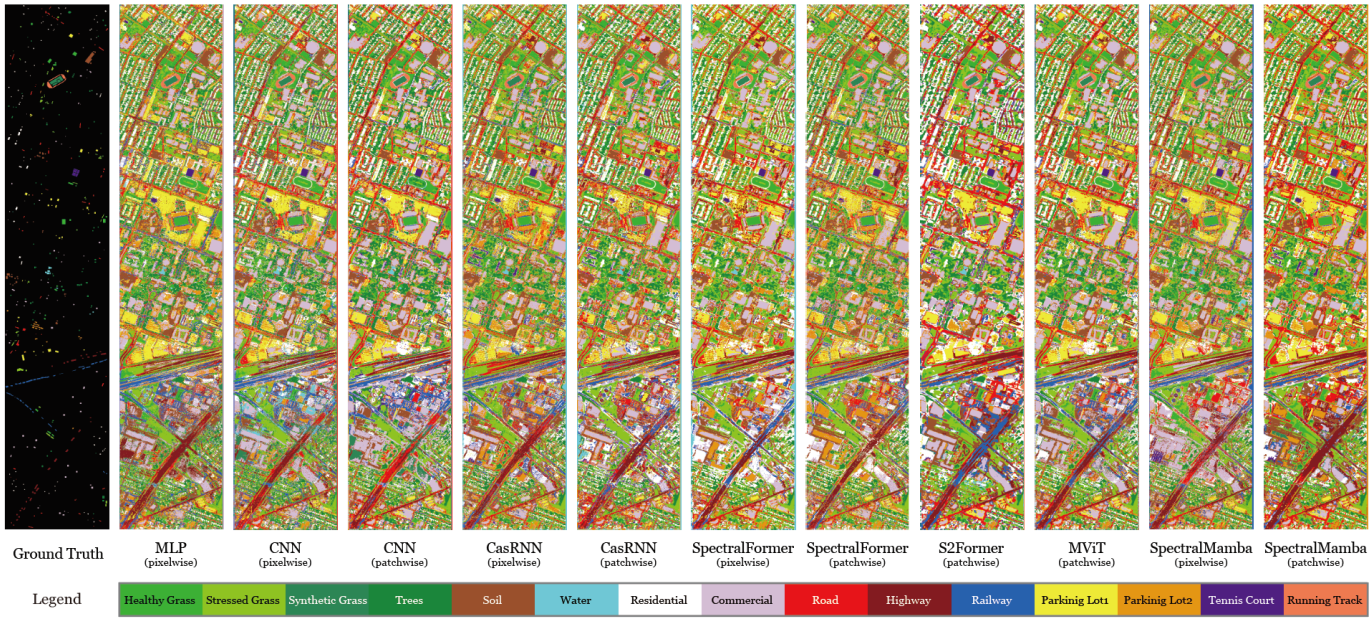


Fig. 5. An illustration of the train/test labels and classification maps obtained by compared methods on the Houston2013 HS dataset.

1) First, the samples selected for training are better evenly distributed in the scene to ameliorate the spectral variability phenomenon that may deteriorate most models.
2) Second, it is important to maintain a moderate train set size to prevent the evaluation that is either too simplistic or too challenging, therefore striking a balance in accurately reflecting the model's performance and generalization ability.
3) Third, rather than pixelwise sampling, image segment-based sampling or labeling tends to be more efficient in progressively constructing the train set to a preset size.
4) Last but not least, class-balanced sampling is always preferable, especially for unseen scenes with no prior knowledge of the class distribution.

Based on the above criteria, we propose the following steps to determine the train and test set for those datasets without benchmark split. The first step is to apply the SLIC superpixel method to over-segment the image into a large number of segments [51], most of which can then be made of the same ground object. The next step is straightforward to randomly collect those homogeneous segments progressively until the budget of training samples for each class is reached. This process not only mimics the manual labeling mode under a fixed manpower budget, but also avoids the interference of outliers that are prone to pointwise sampling. The classwise budgets for the last three HS datasets we adopt are empirically set to 80, 50, and 20, respectively. This method can also be used to organize a separate validation set [52].

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TABLE II
QUANTITATIVE COMPARISON WITH RELATED REPRESENTATIVES ON HOUSTON2013 HS DATASET. THE BEST AND SECOND-BEST OVERALL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	MLP	DCNN [23]		CasRNN [27]		SpectralFormer [32]		S2Former [49]	MViT [31]	SpectralMamba	
Implementation	Pixelwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Patchwise	Patchwise	Pixelwise	Patchwise
Healthy Grass	83.29	84.24	82.24	83.67	82.62	82.81	82.53	94.11	84.14	93.83	**95.92**
Stressed Grass	59.59	68.05	86.84	97.74	98.31	98.68	96.71	98.03	98.50	95.30	**98.87**
Synthetic Grass	98.81	91.09	96.24	**100.00**	96.44	99.60	98.22	97.43	99.21	99.80	98.61
Tree	83.71	80.59	**99.72**	98.20	98.96	98.48	98.20	98.01	99.62	96.50	99.05
Soil	46.12	49.62	85.51	97.73	97.92	96.02	98.77	**99.91**	**99.91**	98.86	**99.91**
Water	93.71	92.31	**95.80**	95.10	95.10	95.10	93.01	95.10	**95.8**	95.10	94.41
Residential	59.51	78.64	78.54	86.75	85.26	85.91	**91.32**	84.98	89.37	82.46	87.03
Commercial	69.42	64.01	82.24	60.21	66.29	50.71	67.43	**83.57**	78.16	77.30	64.86
Road	76.30	78.56	70.73	74.22	71.48	73.56	71.95	73.28	69.31	**82.06**	80.64
Highway	53.38	39.77	50.10	81.37	69.40	88.51	86.29	82.72	85.14	60.33	**98.17**
Railway	81.69	56.45	85.77	88.80	83.02	75.81	81.02	78.27	**89.28**	81.40	80.36
Parking Lot 1	52.83	42.94	62.73	68.01	57.44	65.03	65.90	73.78	80.69	74.26	**81.56**
Parking Lot 2	67.02	57.89	81.40	75.79	84.91	72.63	69.82	**86.67**	89.82	72.63	79.65
Tennis Court	92.31	94.33	99.60	99.60	98.79	99.60	97.17	98.79	98.38	99.19	**100.00**
Running Track	96.19	98.10	94.50	97.89	95.35	98.73	99.15	**99.79**	**99.79**	98.31	98.52
OA (%) ↑	69.94 ±1.61	67.58 ±1.20	80.57 ±1.33	85.21 ±2.09	82.94 ±2.25	83.32 ±2.53	85.27 ±2.11	87.99 ±1.98	88.78 ±2.16	85.64 ±1.84	**89.52** ±1.65
AA (%) ↑	74.26 ±1.55	71.77 ±1.32	83.46 ±1.47	87.01 ±2.22	85.42 ±2.60	85.41 ±2.56	86.50 ±2.20	89.63 ±1.77	90.47 ±1.99	87.16 ±1.21	**90.50** ±1.57
κ ↑	0.6749 ±0.0151	0.6501 ±0.0098	0.7894 ±0.0112	0.8397 ±0.0202	0.8149 ±0.0221	0.8193 ±0.0238	0.8402 ±0.0220	0.8696 ±0.0199	0.8782 ±0.0197	0.8442 ±0.0165	**0.8864** ±0.0170
MACs (M) ↓	**19.64**	26.83	640.09	548.45	549.95	667.04	681.19	264.23	34.95	23.52	36.21
Params (K) ↓	305.94	418.77	1192.96	350.61	353.49	72.56	74.10	86.68	43.38	**14.23**	36.55
Time (Min.) ↓	**0.89**	1.76	3.11	2.01	3.46	3.13	4.36	3.08	1.98	1.63	2.45

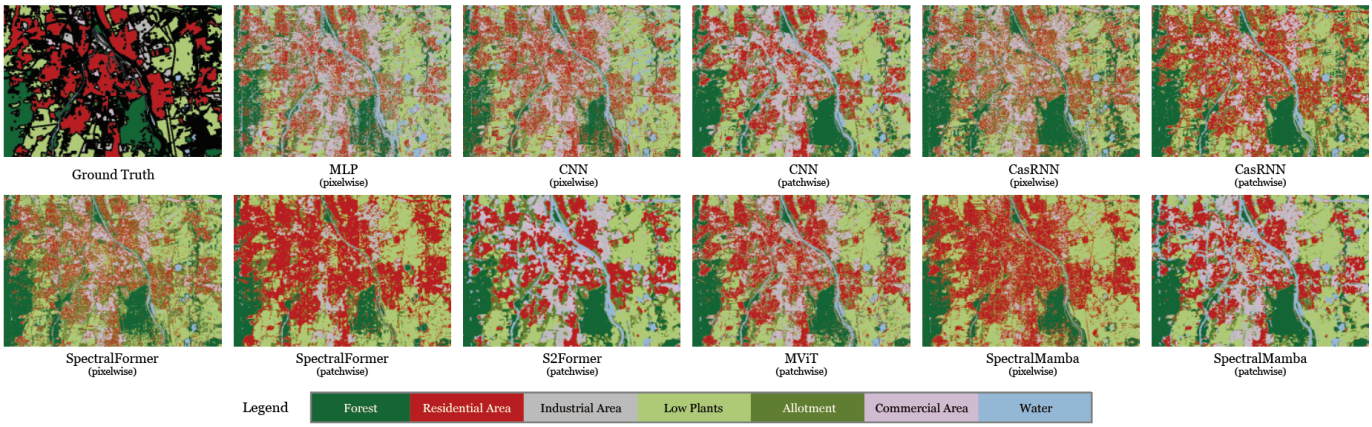


Fig. 6. An illustration of the false-color image, train and test labels, and classification maps obtained by compared methods on the Augsburg HS dataset.

2) Evaluation Metrics: Besides the conventional performance metrics such as classwise accuracy (CA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ), we introduce two more metrics to evaluate the efficacy of different methods, which are multiply-accumulate operations (MACs) and parameters (Params) of each network. Given K classes with n_k test samples in class k and n_{kk} correctly classified, OA is defined as $\text{OA} = \sum_{k=1}^{K} n_{kk} / \sum_{k=1}^{K} n_k$, and AA as the mean of per-class accuracies, i.e., $\text{AA} = \frac{1}{K} \sum_{k=1}^{K} n_{kk}/n_k$. The Kappa coefficient measures inter-rater agreement accounting for chance: $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o equals OA and p_e is the expected agreement by chance. As the name suggests, MACs refer to the number of multiply-accumulate operations during actual network training. In our experiments, we set the batch size for all methods as 64 to fairly compute and compare their MACs for one batch. The less value the MACs and Params take, the less computational resource the corresponding model costs.

3) Compared Methods: The major aim of our experiments is to assess whether the proposed SpectralMamba can be deemed as a revolutionized DL tool for the HS image classification task. Therefore, we select five prevalent types of DL-based solutions for comparison, which are MLP, CNN, RNN-based, spectral Transformer, and spatial-spectral Transformer-based models. The details of our selected competitors are as follows.

1) The MLP contains an input block, two hidden blocks, and a classification head. Each of the first three blocks consists of a fully-connected (FC) layer, a 1-D batch normalization (BN) layer, and a ReLU activation.
2) The CNN contains two convolution blocks and a classification head. The convolution block consists of 1-D or 2-D convolution layers depending on pixelwise or patchwise input [23], corresponding 1-D or 2-D BN layers, and ReLU.
3) The CasRNN is selected to represent the RNN-based

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TABLE III
QUANTITATIVE COMPARISON WITH RELATED REPRESENTATIVES ON AUGSBURG HS DATASET. THE BEST AND SECOND-BEST OVERALL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	MLP	DCNN [23]		CasRNN [27]		SpectralFormer [32]		S2Former [49]	MViT [31]	SpectralMamba	
Implementation	Pixelwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Patchwise	Patchwise	Pixelwise	Patchwise
Forest	13.27	12.15	53.77	94.30	96.31	90.23	92.74	**98.49**	94.79	87.96	97.57
Residential Area	23.61	32.71	35.84	47.53	63.76	45.50	53.58	67.29	**74.66**	57.33	73.99
Industrial Area	34.37	25.40	32.67	30.76	9.71	24.85	**54.65**	42.93	52.40	35.38	51.13
Low Plants	22.90	29.86	38.45	57.23	66.52	71.43	**84.46**	81.53	77.67	77.39	80.95
Allotment	62.63	48.48	80.00	71.11	90.30	82.22	75.35	**92.53**	89.90	77.37	90.51
Commercial Area	42.56	30.22	45.30	45.30	**58.34**	53.04	36.17	56.36	41.85	48.12	48.95
Water	51.83	50.38	61.08	53.42	63.35	49.55	64.67	**78.74**	69.57	61.90	75.57
OA (%) ↑	23.26 ±1.06	28.20 ±1.11	40.63 ±1.20	58.35 ±1.53	67.77 ±1.83	61.62 ±1.94	71.03 ±2.09	76.56 ±2.05	77.44 ±1.98	68.49 ±1.36	**78.98** ±1.72
AA (%) ↑	35.88 ±1.22	32.75 ±1.40	49.59 ±1.56	57.09 ±1.77	64.04 ±1.88	59.54 ±1.80	65.95 ±1.83	73.98 ±1.92	71.55 ±1.66	63.64 ±1.50	**74.09** ±1.80
κ ↑	0.0942 ±0.0067	0.1094 ±0.0101	0.2692 ±0.0122	0.4645 ±0.0108	0.5673 ±0.0153	0.5018 ±0.0165	0.6157 ±0.0162	0.6894 ±0.0159	0.6950 ±0.0170	0.5864 ±0.0146	**0.7170** ±0.0138
MACs (M) ↓	**20.10**	41.74	1038.34	685.38	687.24	832.60	850.29	343.31	44.23	30.96	50.55
Params (K) ↓	313.10	651.61	1861.63	348.55	352.15	72.04	73.58	94.5	56.97	**13.56**	47.94
Time (Min.) ↓	**1.03**	2.27	5.00	4.16	5.11	4.34	6.75	4.04	2.05	2.01	3.98

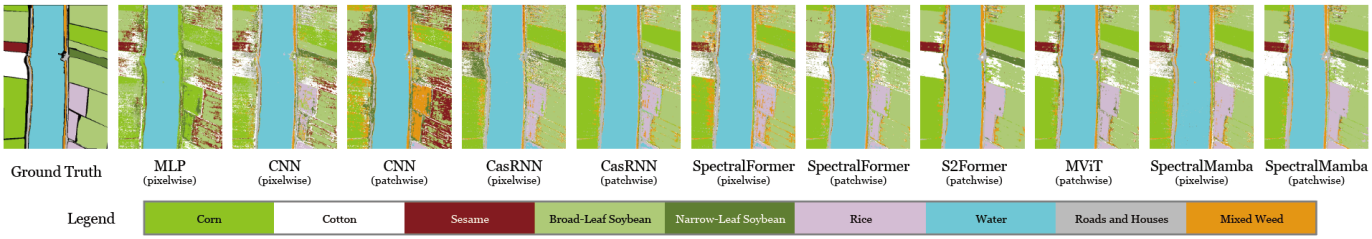


Fig. 7. An illustration of the false-color image, train and test labels, and classification maps obtained by compared methods on the Longkou HS dataset.

models [27]. It has pixelwise and patchwise versions. The pixelwise CasRNN mainly contains cascaded gated recurrent units, while the patchwise one additionally introduces two separable convolution blocks and a max-pooling layer before the pixelwise CasRNN[1].

4) The pixelwise SpectralFormer and patchwise SpectralFormer are selected to represent the Transformer-based models [32]. The former one is equipped with the groupwise spectral embedding to enhance local spectral details while the latter adds an FC layer to encode the spatial information from flattened image patches[2].
5) The patchwise SpectralFormer takes plain vectorization on patches. Therefore, a more dedicated Transformer-based that considers spatial-spectral transformed feature has been taken into comparison, termed as S2Former [49].
6) To further validate the efficiency and robustness of the proposed method against state-of-the-art Transformers, we selected the masked ViT (MViT) [31], which is the first lightweight HS image classification method based on masking mechanism[3].

4) Implementation Details: All of our experiments are conducted mainly based on the PyTorch framework using a workstation with an Nvidia GeForce GTX 3080 GPU card. We tuned the learning rate and weight decay hyperparameters via a rough grid search on intervals of $\{10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}\}$. The number of epochs is set to 500. A StepLR scheduler is adopted to shrink the learning rate by multiplying a factor of 0.9 for each 20 epochs. All statistical results are computed based on 5 repeated experiments using different random seeds, hoping to eliminate the effect caused by random initialization and optimization. To ensure a fair comparison, all baseline methods were retrained from scratch under exactly the same train/test splits using the same set of random seeds (0–4). No pre-trained weights or external data were used for any method.

Another commonly overlooked issue is information leakage, that is, more testing samples will be seen as the patch size of training samples increases [53]. To mitigate the effect caused by this issue, we set the patch size for all patchwise models as 3. We claim that this setup is enough for assessment from the perspective of tendency, which also consumes less energy and is aligned with the Sustainable Development Goals. We additionally measured peak GPU memory consumption using `torch.cuda.max_memory_allocated()` during training with batch size 64 on our GTX 3080. SpectralMamba (patchwise) consumes 3.2–4.8 GB depending on the dataset, which is 30–45% less than the pathwise version of SpectralFormer and comparable to MViT.

C. Results and Analysis

We summarize the quantitative results of all compared methods on six metrics, i.e., performance metrics as CA, OA, AA, and κ, and efficacy metrics as MACs and Params, in Tables II to V for Houston2013, Augsburg, Longkou, and Botswana datasets, respectively. The classification maps of full scenes are correspondingly visualized in Figs. 5 to 8.

[1] https://github.com/RenlongHang/CasRNN
[2] https://github.com/danfenghong/IEEE_TGRS_SpectralFormer
[3] https://github.com/swiftest/MViT

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TABLE IV
QUANTITATIVE COMPARISON WITH RELATED REPRESENTATIVES ON LONGKOU HS DATASET. THE BEST AND SECOND-BEST OVERALL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	MLP	DCNN [23]		CasRNN [27]		SpectralFormer [32]		S2Former [49]	MViT [31]	SpectralMamba	
Implementation	Pixelwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Patchwise	Patchwise	Pixelwise	Patchwise
Corn	37.36	12.68	4.64	89.38	95.98	85.47	85.47	**97.84**	94.96	91.60	97.48
Cotton	28.16	25.12	33.29	30.50	70.46	69.53	69.53	**97.44**	94.33	74.49	95.01
Sesame	29.63	30.50	39.13	51.63	65.91	60.36	60.36	82.36	81.26	78.17	**92.17**
Broad-Leaf Soybean	30.95	32.32	47.81	73.85	77.15	73.44	73.44	80.4	81.29	79.27	**82.87**
Narrow-Leaf Soybean	32.63	25.24	66.49	76.42	86.45	83.26	83.26	92.29	**94.02**	90.85	90.68
Rice	0.00	22.29	13.20	80.77	83.62	79.67	79.67	**98.33**	94.23	95.41	98.29
Water	7.47	6.17	80.98	99.83	**99.99**	**99.99**	**99.99**	99.97	99.97	98.66	99.60
Roads and Houses	0.00	13.53	7.27	79.86	86.17	77.90	77.90	81.91	86.52	61.76	**88.81**
Mixed Weed	0.00	23.99	33.45	48.05	54.63	53.13	53.13	**83.33**	72.83	68.75	73.74
OA (%) ↑	20.57 ±1.03	18.51 ±1.11	47.30 ±1.20	82.92 ±1.37	87.69 ±1.44	84.03 ±1.42	88.86 ±1.97	91.90 ±1.48	91.23 ±1.16	87.80 ±2.02	**92.48** ±1.33
AA (%) ↑	18.47 ±0.94	21.32 ±1.08	36.25 ±1.23	70.03 ±1.42	80.04 ±1.50	75.86 ±1.60	82.29 ±2.07	90.43 ±1.72	88.82 ±1.18	82.11 ±1.90	**90.96** ±1.59
κ ↑	0.0575 ±0.0043	0.0693 ±0.0055	0.3558 ±0.0142	0.7815 ±0.0169	0.8417 ±0.0167	0.7964 ±0.0158	0.8566 ±0.0189	0.8955 ±0.0140	0.8869 ±0.0099	0.8432 ±0.0198	**0.9028** ±0.0136
MACs (M) ↓	**21.61**	93.74	2333.70	954.68	957.48	1276.52	1303.65	608.66	80.97	52.79	93.18
Params (K) ↓	336.65	1499.14	4186.11	349.07	354.47	72.17	73.71	82.28	63.38	**18.68**	94.55
Time (Min.) ↓	**2.13**	4.97	6.08	5.85	7.69	7.30	9.68	6.39	5.66	5.82	7.44

From the tables, we can draw several conclusions that are consistent on four datasets. Patchwise implementations generally tend to produce better performance than pixelwise ones by exploitation of spatial information (except for CasRNN on the Houston2013 dataset, which is probably caused by the unstable training issue). In specific, pixelwise MLP and pixelwise CNN involve fewer computations than CasRNN and SpectralFormer. Still, patchwise CNN encounters an evident explosion in terms of MACs due to its use of inseparable 2-D convolution. If we examine it further through the lens of computational efficiency, although SpectralFormer owns much fewer parameters than those of CasRNN, its self-attention operations result in higher MACs, particularly for dealing with longer sequences in the middle two datasets. Two most competitive Transformer-based methods, i.e., S2Former and MViT, can surpass the performance of SpectralFormer, while MViT enjoys further computational efficiency than the S2Former. Most importantly, our SpectralMamba unsurprisingly achieves the best CAs in most classes, the best OA, AA, and κ, and meanwhile the best Params, and the MACs that are only second to those of the simplistic MLP with a negligible margin.

Some other interesting observations are also worth noting. If we set the quantitative tendency on the Houston2013 dataset as the reference, both the conventional pixelwise MLP and pixelwise CNN behave contrarily on Botswana and the other two datasets. This could possibly be explained by that the train and test sample distributions are much closer to each other on the Botswana dataset than on the other two. Patchwise CNN raises the performance of pixelwise CNN by 12.43% and 28.79% in terms of OA on the middle two datasets, but still at a low level. CasRNN, SpectralFormer, and our SpectralMamba, no matter whether pixelwise or patchwise implementation, consistently show a stable classification performance at a desirable level, which verifies the importance and superiority of their sequential modeling capability. Furthermore, our SpectralMamba successfully addresses the computational weaknesses that lie in the former two prevalent sequence models by leveraging appropriate spatial-spectral merging, sequential scanning, and state space modeling.

We can draw clues from the visualization results that exhibit trends close to those of the above-mentioned quantitative results. For example, our SpectralMamba offers more reliable predictions for categories such as *road* and *highway* in the Houston2013 scene. As for the Augsburg scene, although CasRNN and SpectralFormer also show *forest* as good as ours, they typically confuse *residential area* with other classes. Moreover, the trajectory of the river and the shape of paddy fields, i.e., *water* category, can be clearly depicted by our SpectralMamba. The superiority of our methods is more obvious in the Longkou scene. Take *rice* area as an instance, in contrast to other methods that are easy to mix it with *water* or *mixed weed*, ours discover a region that is most regular and homogeneous used to plant *rice*. Although the labels in the Botswana scene are more sparse, we can still tell that our patchwise SpectralMamba generates classification maps with less salt and pepper noises-like patterns that are more likely in accord with the real distribution of ground objects.

To further validate the statistical significance of our results, we conducted paired Wilcoxon signed-rank tests between SpectralMamba (patchwise) and the second-best method (MViT) across all four datasets, yielding $p < 0.05$ for OA on all datasets, confirming that the improvements are statistically significant. A closer look at the per-class accuracies reported in Tables II–V further reveals SpectralMamba's advantage in reducing spectral confusion among challenging class pairs. For example, on the Houston2013 dataset, SpectralMamba raises the accuracy of *Highway* from 85.14% (MViT) to 98.17% and *Road* from 69.31% to 80.64%, two categories that are easily confused due to similar spectral profiles under different spatial contexts. Analogous improvements can be observed for *Residential* vs. *Commercial*. These gains are attributable to the GSSM module's ability to generate spatially-adaptive spectral representations that better distinguish spectrally similar but contextually different materials.

D. Ablation Studies

1) Patch Size: We first investigate the impact of patch size on the performance and efficiency of our SpectralMamba.

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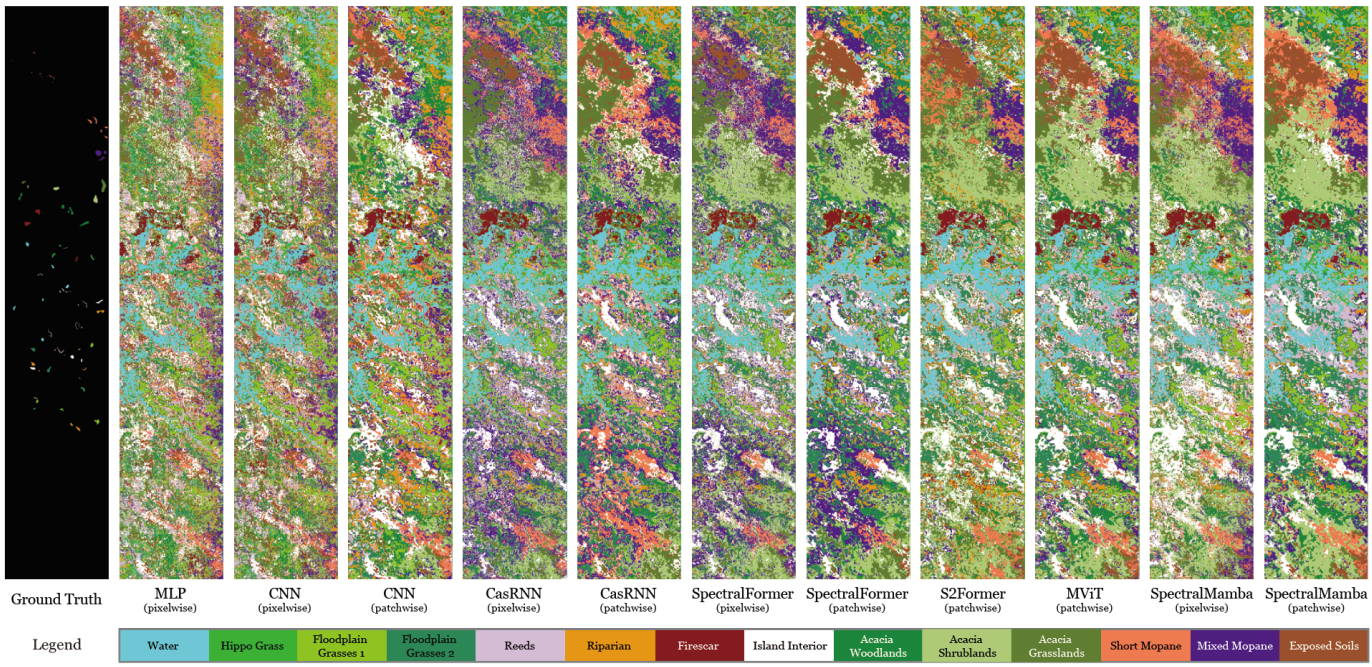


Fig. 8. An illustration of the false-color image, train and test labels, and classification maps obtained by compared methods on the Botswana HS dataset.

TABLE V
QUANTITATIVE COMPARISON WITH RELATED REPRESENTATIVES ON BOTSWANA HS DATASET. THE BEST AND SECOND-BEST OVERALL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

Method	MLP	DCNN [23]		CasRNN [27]		SpectralFormer [32]		S2Former [49]	MViT [31]	SpectralMamba	
Implementation	Pixelwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Pixelwise	Patchwise	Patchwise	Patchwise	Pixelwise	Patchwise
Water	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100**	**100**	**100.00**	**100.00**
Hippo Grass	**100.00**	**100.00**	**100.00**	97.53	97.53	**100.00**	98.77	**100**	**100**	95.06	**100.00**
Floodplain Grasses 1	94.78	98.26	**100.00**	81.30	96.52	90.87	95.65	**100**	99.57	95.65	**100.00**
Floodplain Grasses 2	96.86	96.86	94.76	88.48	94.76	93.72	92.15	95.81	95.29	92.15	**97.91**
Reeds	90.36	89.16	90.76	75.90	73.49	79.52	87.55	**93.17**	93.17	86.75	**93.17**
Riparian	71.89	82.73	**97.59**	71.08	68.27	65.86	87.15	92.37	91.5700	77.91	94.78
Firescar	97.49	96.23	**100.00**	96.65	89.96	87.87	99.58	**100**	**100**	97.91	**100.00**
Island Interior	99.45	98.91	**100.00**	91.26	97.81	97.27	**100.00**	99.45	99.45	95.63	**100.00**
Acacia Woodlands	84.98	74.74	90.10	72.35	72.35	74.74	87.71	92.49	91.47	86.35	**96.25**
Acacia Shrublands	95.13	93.36	**97.79**	73.45	74.34	92.48	95.13	**99.12**	96.90	96.02	97.79
Acacia Grasslands	87.37	92.28	**98.25**	85.96	92.28	84.91	97.19	92.98	94.04	85.61	94.04
Short Mopane	96.23	95.60	**100.00**	99.37	**100.00**	96.86	**100.00**	**100**	99.37	91.19	99.37
Mixed Mopane	89.52	91.94	98.39	76.61	89.92	88.71	93.55	98.79	99.19	97.58	**99.60**
Exposed Soils	**100.00**	98.67	98.67	94.67	98.67	98.67	**100.00**	**100**	**100**	**100.00**	**100.00**
OA (%) ↑	91.81 ±1.08	92.21 ±0.92	97.19 ±0.86	84.19 ±2.20	87.14 ±1.55	87.44 ±1.71	94.55 ±1.31	96.89 ±1.10	96.58 ±0.99	91.88 ±1.35	**97.66** ±0.94
AA (%) ↑	93.15 ±1.12	93.48 ±0.97	97.59 ±0.71	86.05 ±1.96	88.99 ±1.56	89.39 ±1.87	95.32 ±1.42	97.44 ±1.08	97.14 ±1.02	92.70 ±1.21	**98.06** ±0.68
κ ↑	0.9112 ±0.0097	0.9156 ±0.0101	0.9695 ±0.0082	0.8287 ±0.0121	0.8606 ±0.0119	0.8639 ±0.0133	0.9409 ±0.0107	0.9662 ±0.0099	0.9629 ±0.0098	0.9119 ±0.0118	**0.9747** ±0.0080
MACs (M) ↓	**19.64**	27.19	658.12	570.80	572.31	671.63	685.89	237.83	38.43	23.39	36.25
Params (K) ↓	305.93	424.43	1209.34	350.35	353.25	72.49	74.03	75.61	56.08	**12.33**	34.96
Time (Min.) ↓	**0.21**	0.29	0.80	1.01	1.47	2.03	3.56	1.53	0.33	0.29	0.61

We conduct experiments by varying the patch size from 3 to 11 on the Houston2013 dataset. The quantitative results on OA, MACs, and parameter numbers are demonstrated in Fig. 9. From the figure we can observe an obvious performance decline after a steady increase, while the computational metrics are increasing accordingly. This could be partially explained by the fact that the train and test split is well separated in this dataset, therefore simply increasing the patch size may bring information gain but can not guarantee a constant performance improvement. Note that thanks to our lightweight design of the first-stage spatial-spectral feature mining module, i.e., GSSM, the increase number of parameters is much more negligible compared to that of computations. Hence we propose to set the patch size parameter as 3 throughout our experiments by weighing the trade-off between model performance and efficiency.

2) Module Effectivity: Then we conduct ablation experiments on three key components in our SpectralMamba, that are, GSSM, PSS, and Mamba. We make a quantitative comparison by removing one or two constituents are summarized in Table VI. In most cases, an evident relative improvement can be observed on the basis of a single use of Mamba under pixelwise implementation. For example, our proposed scanning strategy brings nearly 1% and 3% accuracy improvement

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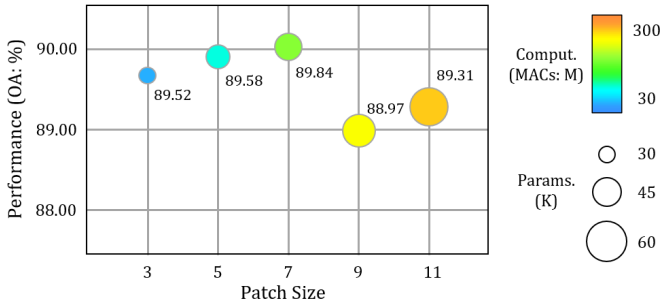


Fig. 9. Ablation results of our SpectralMamba on Houston2013 HS Dataset by varying the patch size of model input.

TABLE VI
QUANTITATIVE RESULTS OF ABLATION BY SWITCHING ON AND OFF THREE KEY MODULES IN THE PROPOSED SPECTRALMAMBA ON HOUSTON2013 HS DATASET. THE RELATIVE IMPROVEMENT AND DEGRADATION ARE SHOWN IN CYAN AND RED, RESPECTIVELY.

Module	Implementation					
	Pixelwise		Patchwise			
GSSM	-	-	✗	✓	✓	✓
PSS	✗	✓	✓	✗	✓	✓
Mamba	✓	✓	✓	✓	✗	✓
OA (%)	84.62	85.59 (+0.97)	86.74 (+2.12)	85.73 (+1.11)	84.40 (-0.22)	89.52 (+4.90)
AA (%)	85.56	87.23 (+1.67)	88.40 (+2.84)	87.55 (+1.99)	86.57 (+1.01)	90.50 (+4.94)
κ	0.8330	0.8438 (+0.0108)	0.8561 (+0.0231)	0.8451 (+0.0121)	0.8308 (-0.0008)	0.8864 (+0.0534)
MACs (M)	43.21	23.25 (-19.96)	23.60 (-19.61)	55.90 (+12.69)	12.73 (-30.48)	36.21 (-7.00)
Params (K)	66.17	14.23 (-51.94)	14.24 (-51.93)	88.49 (+22.32)	22.75 (-43.42)	36.55 (-29.62)

under pixelwise and patchwise settings, respectively. What's more, it can lead to 20M and 50K decreases in MACs and Params respectively, compared to the scenarios where only this module is switched off. We also observe a significant improvement in terms of 2.78% OA and 2.10% AA by activating the GSSM module in patchwise case. Note that the Mamba can still be deemed the most valuable module as more than 5% OA decrease occurs when removing it.

To verify the generality of each module, we extend the ablation to the remaining three datasets (Augsburg, Longkou, and Botswana) and further examine the contribution of the Mamba (S6) module by replacing it with a two-layer MLP and a 1D convolutional block, respectively. The results are summarized in Table VII. The findings are consistent across all datasets: PSS brings 1–4% OA improvement while reducing computation; GSSM contributes 1.5–3% OA in the patchwise setting; and replacing Mamba with MLP or 1D-Conv degrades OA by 3.0–6.1%, confirming the essential role of input-dependent state evolution in S6.

3) Merging Style: To further validate the design choice of our GSSM module, we compare the depthwise-separable (DW+PW) convolutional gating against several alternative gating mechanisms: (1) a single standard 3×3 convolution, (2) a squeeze-and-excitation (SE) block [45] with reduction ratio $r = 4$, (3) a convolutional block attention module (CBAM) [54], and (4) no gating (baseline). As summarized in Table VIII, our DW+PW design consistently achieves the best OA across all four datasets. The SE block improves over the no-gating baseline but underperforms GSSM by 1.2–2.1%, as it only provides channel-wise attention without spatial adaptivity. The standard 3×3 convolution performs competitively but remains 0.3–0.8% below our DW+PW design. These results collectively confirm that the effectiveness of GSSM is robust to architectural choices, while the depthwise-separable design provides the best accuracy among all tested gating mechanisms.

TABLE VII
EXTENDED MODULE ABLATION ACROSS DATASETS AND MAMBA REPLACEMENT STUDY. OA (%) IS REPORTED FOR PATCHWISE SPECTRALMAMBA.

Configuration	HU	AG	LK	BW
Full (GSSM+PSS+Mamba)	89.52	78.98	92.48	97.66
w/o GSSM (PSS+Mamba)	86.74	76.55	91.03	96.51
w/o PSS (GSSM+Mamba)	85.73	75.58	90.23	94.37
w/o Mamba (GSSM+PSS)	84.40	75.75	89.90	95.33
Mamba $\rightarrow$ MLP	83.28	74.38	88.76	94.33
Mamba $\rightarrow$ 1D-Conv	84.10	74.33	89.01	94.46

TABLE VIII
QUANTITATIVE COMPARISON OF DIFFERENT GATING MECHANISMS IN THE GSSM MODULE. OA (%) IS REPORTED FOR ALL FOUR DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD.

Gating Mechanism	HU	AG	LK	BW
None	86.74	75.47	89.68	95.23
SE [45]	87.43	76.88	90.82	96.46
CBAM	88.19	77.63	91.45	96.89
Std. Conv	88.93	78.41	91.69	97.28
DW+PW (Ours)	**89.52**	**78.98**	**92.48**	**97.66**

4) Scanning Pieces: The other ablation experiment worth conducting is to investigate the impact of varying the number of scanning pieces, i.e., R. The quantitative results are illustrated in Fig. 10 by setting R to values from 2 to 8 with an interval of 2. From the figure, we can observe that the baseline without using a scanning strategy can still attain an acceptable performance, while its computation and parameter scales are much higher than those using scanning. Specifically, in both pixelwise and patchwise cases, the MACs and Params metrics simultaneously show an initial decrease followed by an increase as R increases, in contrast to the trend for performance. The underlying reason behind this phenomenon lies in that too many pieces inevitably widen the networks for SSM and increase the difficulty in pattern recognition from resulting spectral sequences with short lengths. Fortunately, by trading off all metrics, we can find the optimal $R = 6$ for the Houston2013 dataset, whose computation and parameters are maximally 46% and 79% less than w/o scanning while bringing 1% to 4% improvements in OA approximately. We further extend this analysis to Augsburg, Longkou, and Botswana to verify whether the same R generalizes across sensors. As shown in Table IX, $R = 6$ consistently yields the best or near-best OA across all four datasets, confirming that this value can be adopted as a universal default without per-dataset tuning.

5) Generalization Ability: To explore the robustness and generalizability of our proposed spectral feature extraction backbone in complex scenarios, we conducted two additional cross-scene classification experiments. First, we targeted the

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TABLE IX
OA (%) OF PATCHWISE SPECTRALMAMBA WITH VARYING R ACROSS DATASETS. THE BEST RESULT PER DATASET IS IN BOLD.

R	2	4	6	8	w/o PSS
HU (L=144)	83.35	87.10	**89.52**	85.34	85.73
AG (L=180)	76.43	77.52	**78.98**	78.03	74.55
LK (L=270)	90.11	92.03	**92.48**	91.51	89.89
BW (L=145)	95.35	94.47	97.66	**97.71**	93.28

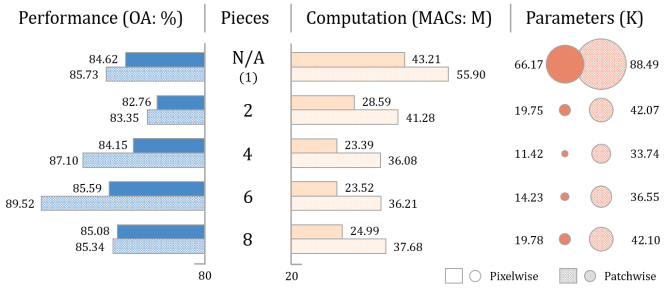


Fig. 10. Ablation results of our SpectralMamba on Houston2013 HS Dataset by varying the number of pieces for the proposed piece-wise sequential scanning.

TABLE X
QUANTITATIVE RESULTS OF GENERALIZATION ABILITY TEST ON TWO DIFFERENT CROSS-SCENE SETTINGS.

Dataset	Pavia University		LN-01	
Method	SpectralFormer	SpectralMamba	SpectralFormer	SpectralMamba
OA	34.64 ±1.93	38.36 ±1.77	87.58 ±2.01	91.09 ±1.58
AA	39.05 ±2.08	45.90 ±1.97	86.53 ±1.78	89.39 ±1.61
κ	0.3785 ±0.0222	0.4437 ±0.0181	0.8421 ±0.0158	0.8828 ±0.0134

TABLE XI
QUANTITATIVE COMPARISON WITH DOMAIN ADAPTATION ON THE PAVIA CENTER → UNIVERSITY CROSS-SCENE SETTING.

	SFormer	SpectralFormer-D	SpectralMamba	SpectralMamba-D
OA	34.64	42.87	38.36	**46.53**
AA	39.05	48.23	45.90	**52.18**
κ	0.3785	0.4892	0.4437	**0.5301**

widely-studied Pavia datasets, including the University of Pavia and Pavia Center. By merging and deleting classes, we curated a 7-class labeled dataset for the cross-scene task[4]. Simultaneously, we performed band matching on the hyperspectral data from these two scenes to obtain data with 102 identical bands. We used the Pavia Center dataset for training and the University of Pavia dataset for prediction. We ran 5 repeated experiments, and the results are compiled in Table X. As can be seen, the results for both our method and the baseline SpectralFormer are only around 40%. The reason behind this phenomenon lies in that these two datasets have significant distribution discrepancies even for the same land cover types, possibly due to different preprocessing methods. Furthermore, our method, as a conventional HS image classification approach, lacks a specific design for cross-domain distribution alignment. Despite this, the consistent improvement over SpectralFormer still demonstrates the advantage of our method in feature extraction.

To further contextualize the Pavia cross-scene results, we additionally compare with domain-adapted versions of both backbones [55]. As shown in Table XI, adding domain adaptation (-D) improves the OA for both methods, confirming that the 35–38% range is indeed expected for vanilla classification methods under severe domain shift. Importantly, SpectralMamba consistently provides a better starting point than SpectralFormer even with domain adaptation (+3.66% OA), demonstrating the superior feature extraction capability of our backbone. We note that designing dedicated cross-scene methods is beyond the scope of this work, but these results show SpectralMamba can benefit from future integration with domain adaptation techniques.

A closer inspection of per-class performance on the Pavia cross-scene experiment reveals that the most confusing class pairs are *Meadows* vs. *Bare Soil* (sharing similar spectral profiles in the VNIR range with moisture-dependent differences) and *Asphalt* vs. *Bricks* (spectrally similar urban materials). These failure patterns are consistent with the well-documented domain shift between the two scenes and are not unique to our method.

Additionally, we adopted a relatively more reasonable experimental setup using the LN-01 dataset, which was acquired by China's advanced ZY1-02D satellite hyperspectral sensor[5]. We used samples from the upper 16.7% of the same image for training and the rest for testing. The classification results on 10 land cover classes show that, even compared to the non-completely disjoint train/test split setting in the previous comparative experiments section, our method still performs exceptionally well, achieving an OA exceeding 90%. This indicates its potential for practical, operational applications.

V. CONCLUSION

The issues of spectral redundancy and spectral variability have long been plaguing humans from precisely sensing and perceiving the world via HS imaging. Existing intelligent methods for solving the HS image classification task, based on either conventional CNNs or trending sequence models like RNNs and Transformers, can hardly address these issues simultaneously and efficiently. Building upon the foundations laid by recent advances in SSMs, we establish the first HS data-oriented deep SSM model, i.e., SpectralMamba, by elaborately proposing PSS and GSSM to ease the sequentiality learning in the state domain and rectify the spectrum in the spatial-spectral domain, respectively. SpectralMamba enjoys a surprisingly simplified and lightweight architecture and achieves credibly superior HS image classification performance in most aspects. In the future, we will endeavor to unveil its potential under more practical scenarios, such as HS data generalized and precise interpretation on basis of foundation model or fast response under resource-constrained conditions.

[4]https://github.com/YuxiangZhang-BIT/Data-CSHSI

[5]https://github.com/Welcome-to-LISA/Liao-Ning-Datasets

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